

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool : Case Study

Weightage: 10%

QUESTIONS:

Total Marks: 10

Scenario: Web Development Standards Implementation

A startup is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

- A. `<media src="video.mp4"></media>`
- B. `<video src="video.mp4" controls></video>`
- C. `<movie src="video.mp4"></movie>`
- D. `<vid src="video.mp4"></vid>`

Snippet 2: Form Submission Methods

- A. GET
- B. POST

- C. PUT
- D. DELETE

Snippet 3: Navigation Elements

- A. <div>
- B. <nav>
- C. <section>
- D. <header>

Snippet 4: XML Syntax

- A. <note><to>User</to><from>Admin</from>
- B. <note><to>User</to><from>Admin</from></note>
- C. <note><to>User</from></note>
- D. <note to="User"></note>

Snippet 5: Web Communication Protocols

- A. FTP
- B. HTTP
- C. SMTP
- D. SNMP

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

Code of Conduct:

*I **E.HEMANTH REDDY(192372098)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:E.HEMANTH REDDY

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding ; clearly identifies all	Good understanding with most issues	Basic understanding; some issues missed	Poor understanding; problem not identified
Application of Concepts	25	Effectively applies relevant concepts to analyze the	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

LOGIN

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Assessment Portal

CO-1: Internet Programming & Web Standards Case Study

Student Full Name

CHAKRADHAR

Register Number / Roll No

192311439

☒ I certify that this submission is my original work and that I adhere strictly to academic integrity guidelines specified for this assessment.

Authenticate & Start Case Study →

Assessment Tool 3: Case Study • Weightage: 10% • Total Marks: 10

INTERFACE:

 **SIMATS Internet Programming Studio**
Course Code: CSA43 • Outcome: CO-1 (L2)

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Overview

1 Video Tag

2 Form Method

3 Navigation Tag

4 XML Syntax

5 Web Protocols

Final Report

Scenario: Web Development Standards Implementation

Case Study Assessment Tool 3 • Computer Science & Engineering

Start Snippet 1 →

Case Context & Team Observations

A startup is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols. The following observations were made:

Observation 1:
The team needs to embed a promotional video on the homepage.

Observation 2:
A registration form must securely send user data to the server.

Observation 3:
The navigation menu must follow semantic HTML standards.

Observation 4:
XML configuration files are being used but contain syntax errors.

Observation 5:
The application must communicate over the correct web protocol.

QUESTION 1:

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Snippet 1: Video Embedding Options

Observation 1: Select the correct HTML tag for video embedding and justify your answer.

SNIPPET 1 OF 5

Select HTML Tag Option

Option A: `<media src="video.mp4"></media>` Click to Select

Option B: `<video src="video.mp4" controls></video>` Click to Select

Option C: `<movie src="video.mp4"></movie>` Click to Select

Option D: `<vid src="video.mp4"></vid>` Click to Select

Live HTML5 Media Player Testbed

Demonstrating the selected standard `<video>` element in real-time.



```
<video src="video.mp4" controls></video>
```

QUESTION 2:

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Snippet 2: Form Submission Methods

Observation 2: Identify the appropriate HTTP method for form submission and explain why.

SNIPPET 2 OF 5

Select HTTP Method Option

Option A: GET Click to Select

Option B: POST Click to Select

Option C: PUT Click to Select

Option D: DELETE Click to Select

HTTP POST Request Body Inspector

Registration Payload Sample

```
{  
  "name": "Student User",  
  "regNumber": "211561001",  
  "method": "POST",  
  "dataBodySecured": true  
}
```

Simulate HTTP POST Payload Submission

```
// Click button above to simulate HTTP request
```

QUESTION 3:

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Snippet 3: Navigation Elements

Observation 3: Choose the proper semantic element for navigation and justify its usage.

SNIPPET 3 OF 5

Select Navigation Tag Option

Option A: <div>Click to Select

Option B: <nav>Click to Select

Option C: <section>Click to Select

Option D: <header>Click to Select

Semantic DOM Structure Visualizer

<header> Site Branding & Title Bar </header>

★ <nav> Main Navigation Menu Links (<a> </nav>

<main> Page Core Content Section

<footer> Academic Disclaimer & Signature </footer>

← Snippet 2

Proceed to Snippet 4 →

QUESTION 4:

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Snippet 4: XML Syntax Validation

Observation 4: Identify the correct XML syntax and explain the error in other options.

SNIPPET 4 OF 5

Select XML Syntax Option

Option A: <note><to>User</to><from>Admin</from>Click to Select

Option B: <note><to>User</to><from>Admin</from></note>Click to Select

Option C: <note><to>User</from></note>Click to Select

Option D: <note to="User"></note>Click to Select

Live DOMParser XML Validator

```
<note>
  <to>User</to>
  <from>Admin</from>
</note>
```

Validate XML Syntax

✔ XML Document is Well-Formed and Valid!

QUESTION 5:

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Overview1 Video Tag2 Form Method3 Navigation Tag4 XML Syntax5 Web ProtocolsFinal Report

Snippet 5: Web Communication Protocols

Observation 5: Choose the correct protocol for web communication and justify your choice.

SNIPPET 5 OF 5

Select Web Protocol Option

Option A: FTPClick to Select

Option B: HTTPClick to Select

Option C: SMTPClick to Select

Option D: SNMPClick to Select

Protocol Suite Comparison Matrix

Protocol	Primary Role	Web App Fit
HTTP / HTTPS	Web document delivery & APIs	✔ Ideal Choice
FTP	Bulk file transfer	✘ Inappropriate
SMTP	Electronic mail delivery	✘ Mail only
SNMP	Network management	✘ Admin only

RESULT: We Developed well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)